

HR Attrition Analytics Analysis Report

1. Project Overview

This project analyzes employee attrition using the IBM HR Analytics dataset. The workflow combines Python for data preparation and exploratory analysis, SQL for answering business questions, and Power BI for building an interactive dashboard. The objective is to identify workforce trends, monitor employee attrition, and provide actionable insights that support data-driven HR decision making.

2. Business Objective

- Analyze employee attrition across departments, job roles, gender, age groups, and experience.
- Monitor key HR KPIs through an interactive dashboard.
- Identify workforce segments with higher attrition risk.
- Support HR managers with actionable recommendations.

3. Dataset Summary

Dataset: **IBM HR Analytics Employee Attrition**

Rows: 1,470

Columns: 31

The dataset contains employee demographics, compensation, job information, satisfaction metrics, overtime, experience, and attrition status.

Data Quality Summary:

- Checked for missing values.
- Verified duplicate records.
- Validated data types.
- Prepared dataset for SQL and Power BI analysis.

4. Exploratory Data Analysis using Python

Python was used to inspect, clean, and prepare the dataset before analysis.

- Imported dataset using pandas.
- Performed data inspection using `info()`, `describe()`, and `shape`.
- Checked missing values and duplicates.

- Validated data consistency and data types.
- Prepared the cleaned dataset for SQL analysis and Power BI visualization.

```
[1]: import pandas as pd
import numpy as np

[2]: # Loading the dataset using pandas

df = pd.read_csv(r'D:\data\Employee_Attrition_RawData.csv')

[3]: # Display First 5 Rows

df.head(5)
```

	Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Education	EducationField	EmployeeCount	EmployeeNumber	...	RelationshipSatisfacti
0	41	Yes	Travel_Rarely	1102	Sales	1	2	Life Sciences	1	1	...	
1	49	No	Travel_Frequently	279	Research & Development	8	1	Life Sciences	1	2	...	
2	37	Yes	Travel_Rarely	1373	Research & Development	2	2	Other	1	4	...	
3	33	No	Travel_Frequently	1392	Research & Development	3	4	Life Sciences	1	5	...	

5. Data Analysis using SQL

The following business questions were answered using SQL. Paste the corresponding SQL output screenshot below each question.

Q1. How many employees are there in the organization?

Shows the total workforce in the organization and serves as the baseline for all HR analysis.

Results		Messages	
		Total_Employees	
1		1470	

Q2. How many employees stayed and how many left the organization?

Compares employee retention and turnover to evaluate workforce stability.

Results Messages		
	Employee_Status	Total_Employees
1	Stayed	1233
2	Left	237

Q3. What is the overall employee attrition rate?

Measures the percentage of employees who left the organization and indicates overall retention performance.

Results Messages	
	Attrition_Rate_Percentage
1	16.120000000000

Q4. How many employees work in each department?

Displays workforce distribution across departments to support resource planning.

Results Messages		
	Department	Total_Employees
1	Sales	446
2	Research & Development	961
3	Human Resources	63

Q5. What is the gender distribution of employees?

Shows the gender composition of the workforce and supports diversity analysis.

Results Messages		
	Gender	Total_Employees
1	Male	882
2	Female	588

Q6. What is the average monthly income of employees?

Provides an overview of the organization's average employee salary.

Results Messages	
	Average_Monthly_Income
1	6502

Q7. Which department has the highest employee attrition rate?

Identifies the department with the highest employee turnover for targeted retention efforts.

Results Messages			
	Department	Employees_Left	Attrition_Rate
1	Sales	92	20.630000000000

Q8. Which job roles experience the highest employee attrition?

Highlights job roles experiencing the greatest attrition to identify high-risk positions.

Results Messages			
	JobRole	Employees_Left	Attrition_Rate
1	Sales Representative	33	39.760000000000

Q9. Does working overtime increase employee attrition?

Analyzes whether overtime is associated with higher employee attrition.

Results Messages			
	OverTime	Employees_Left	Attrition_Rate
1	1	127	30.530000000000
2	0	110	10.440000000000

Q10. Does monthly income differ between employees who stayed and left?

Compares average salaries of employees who stayed and those who left.

Results Messages		
	Attrition	Average_Monthly_Income
1	0	6832
2	1	4787

Q11. Which age groups experience the highest employee attrition?

Identifies age groups with higher attrition to support targeted HR strategies.

Results Messages			
	Age_Group	Employees_Left	Attrition_Rate
1	18-25	44	35.770000000000
2	26-35	116	19.140000000000
3	56+	8	17.020000000000
4	46-55	26	11.500000000000
5	36-45	43	9.190000000000

Q12. Does job satisfaction influence employee attrition?

Evaluates the relationship between job satisfaction and employee turnover.

Results Messages			
	JobSatisfaction	Employees_Left	Attrition_Rate
1	1	66	22.840000000000
2	3	73	16.520000000000
3	2	46	16.430000000000
4	4	52	11.330000000000

Q13. Rank employees by monthly income within each department.

Ranks employees by salary within each department to analyze pay distribution.

100 %

Results Messages					
	Department	JobRole	Age	MonthlyIncome	Income_Rank
1	Human Resources	Manager	56	19717	1
2	Human Resources	Manager	47	19658	2
3	Human Resources	Manager	55	19636	3
4	Human Resources	Manager	41	19189	4
5	Human Resources	Manager	41	19141	5
6	Human Resources	Manager	59	18844	6
7	Human Resources	Manager	50	18200	7
8	Human Resources	Manager	54	17328	8
9	Human Resources	Manager	42	16799	9
10	Human Resources	Manager	40	16437	10
11	Human Resources	Manager	51	14026	11

Q14. Find the Top 3 highest-paid employees in each department.

Identifies the top earners in each department for compensation analysis.

Results Messages					
	Department	JobRole	Age	MonthlyIncome	Income_Rank
1	Human Resources	Manager	56	19717	1
2	Human Resources	Manager	47	19658	2
3	Human Resources	Manager	55	19636	3
4	Research & Development	Manager	52	19999	1
5	Research & Development	Research Director	41	19973	2
6	Research & Development	Manager	56	19943	3
7	Sales	Manager	51	19847	1
8	Sales	Manager	52	19845	2
9	Sales	Manager	40	19833	3

Q15. Which employees earn more than their department's average salary?

Highlights employees earning more than their department's average salary.

Results		Messages			
	Department	JobRole	Age	MonthlyIncome	Avg_Department_Salary
1	Human Resources	Manager	54	17328	6654
2	Human Resources	Manager	59	18844	6654
3	Human Resources	Human Resources	34	9950	6654
4	Human Resources	Manager	50	18200	6654
5	Human Resources	Human Resources	54	10725	6654
6	Human Resources	Manager	41	19141	6654
7	Human Resources	Manager	41	19189	6654
8	Human Resources	Human Resources	44	10482	6654
9	Human Resources	Manager	56	19717	6654
10	Human Resources	Manager	51	14026	6654
11	Human Resources	Human Resources	29	7988	6654

Q16. Calculate the average monthly income for each job level.

Shows how average salary changes across different job levels.

Results		Messages	
	JobLevel	Avg_JobLevel	
1	1	2786	
2	2	5502	
3	3	9817	
4	4	15503	
5	5	19191	

Q17. Which departments have an attrition rate above the company average?

Identifies departments with attrition rates exceeding the company average.

Results		Messages	
	Department	Attrition_Rate	
1	Human Resources	19.050000000000	
2	Sales	20.630000000000	

Q18. Categorize employees into Experience Levels using CASE.

Classifies employees by experience level for workforce planning and analysis.

Results		Messages			
	Department	JobRole	Age	TotalWorkingYears	Experience_Level
1	Sales	Sales Executive	41	8	Mid-Level
2	Research & Development	Research Scientist	49	10	Mid-Level
3	Research & Development	Laboratory Technician	37	7	Mid-Level
4	Research & Development	Research Scientist	33	8	Mid-Level
5	Research & Development	Laboratory Technician	27	6	Mid-Level
6	Research & Development	Laboratory Technician	32	8	Mid-Level
7	Research & Development	Laboratory Technician	59	12	Senior
8	Research & Development	Laboratory Technician	30	1	Fresher
9	Research & Development	Manufacturing Director	38	10	Mid-Level
10	Research & Development	Healthcare Representative	36	17	Senior
11	Research & Development	Laboratory Technician	35	6	Mid-Level

Q19. Find employees with the longest tenure in each department.

Identifies the longest-serving employees within each department.

Results		Messages			
	Department	JobRole	Age	YearsAtCompany	Tenure_Rank
1	Human Resources	Manager	51	33	1
2	Research & Development	Healthcare Representative	58	40	1
3	Sales	Sales Executive	58	37	1

Q20. Create an HR Summary Report showing employee count, attrition rate, and average salary by department.

Provides a department-wise summary of employee count, attrition rate, and average salary for quick decision-making.

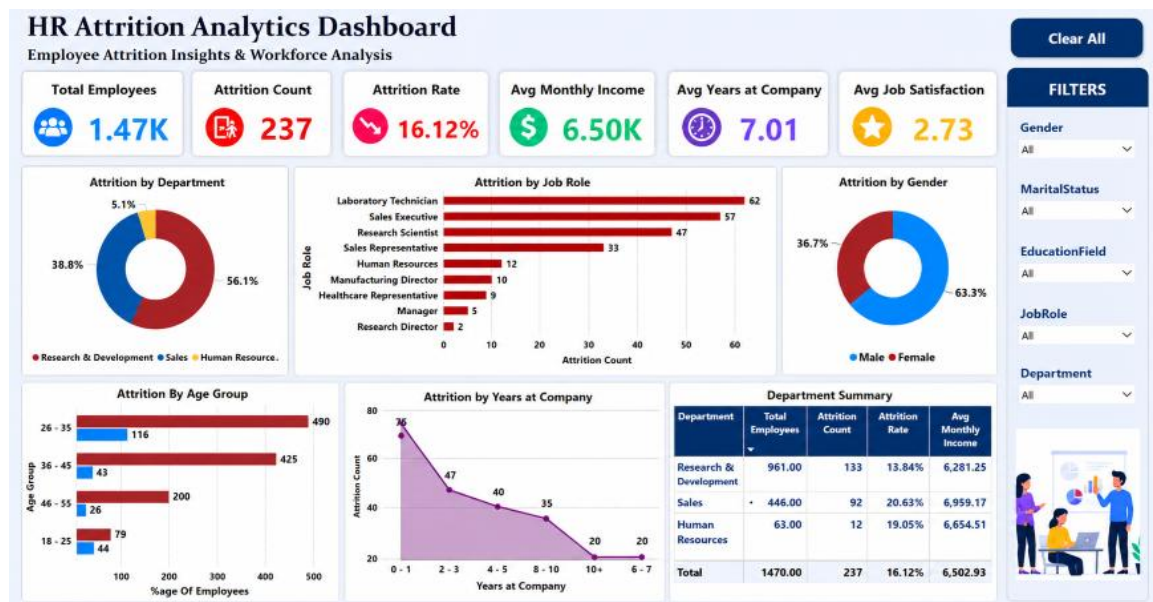
Results		Messages		
	Department	Total_Employees	Attrition_Rate	Average_Salary
1	Human Resources	63	19.050000000000	6654
2	Research & Development	961	13.840000000000	6281
3	Sales	446	20.630000000000	6959

6. Power BI Dashboard

The dashboard provides an interactive overview of workforce metrics through KPI cards, slicers, charts, and summary tables.

- KPI Cards
- Department-wise Attrition

- Job Role Analysis
- Gender Analysis
- Age Group Analysis
- Years at Company Trend
- Department Summary
- Interactive Filters



7. Key Insights

- Overall attrition rate observed across the organization.
- Department with the highest attrition.
- Job role contributing the highest attrition.
- Age group and gender trends.
- Impact of overtime and job satisfaction.
- Salary and tenure observations

8. Business Recommendations

- Strengthen employee onboarding.
- Improve employee engagement and career growth.
- Monitor high-risk departments regularly.

- Review compensation and recognition programs.
- Promote work-life balance initiatives.

9. Conclusion

This project demonstrates an end-to-end HR analytics workflow using Python, SQL, and Power BI. The resulting dashboard and analyses provide meaningful insights into employee attrition and support informed HR decision-making.